

Homework Three Solution– CSE 355

Due: 06 March 2012

Please note that there is more than one way to answer most of these questions. The following only represents a sample solution.

Problem 1: Linz 4.1.13 and 4.2.5

4.1.13: If L is a regular language, prove that $L_1 = \{uv : u \in L, |v| = 2\}$ is also regular.

Let $L_{|2|} = \{v : |v| = 2\}$. Then $L_{|2|}$ is regular since it can be denoted by the regular expression $\Sigma\Sigma$. Here if the alphabet is $\Sigma = \{\sigma_1, \sigma_2, \dots, \sigma_n\}$, then the regular expression Σ is used as shorthand to denote $(\sigma_1 + \sigma_2 + \dots + \sigma_n)$.

Now note that $L_1 = LL_{|2|}$, since $w \in L_1$ iff $w = uv$ with $u \in L$ and $|v| = 2$ iff $w = uv$ with $u \in L$ and $v \in L_{|2|}$ iff $w \in LL_{|2|}$. Since both L and $L_{|2|}$ are regular and by Theorem 4.1 regular languages are closed under concatenation, we conclude the L_1 is regular.

4.2.5: A language is said to be a *palindrome* language if $L = L^R$. Find an algorithm for determining if a given regular language is a palindrome language.

On input L where L is a regular language:

1. Represent L as a DFA, D .
2. Following Theorem 4.2 we construct an NFA, $\hat{N}$ that accepts L^R .
3. Convert $\hat{N}$ into an equivalent DFA, $\hat{D}$. So now $L(\hat{D}) = L^R$.
4. Now we can follow the algorithm in Theorem 4.7 to determine if $L = L^R$, which will give us the result of whether L is a palindrome language.

Note the above algorithm will return that L is a palidrome language iff $L = L^R$, which was the condition we had to check.

Problem 2: Linz 4.3.6 and 4.3.7

4.3.6: Determine whether or not the following languages are regular.

(a) $L = \{a^n b^n : n \geq 1\} \cup \{a^n b^m : n \geq 1, m \geq 1\}$.

Let $L_1 = \{a^n b^n : n \geq 1\}$ and $L_2 = \{a^n b^m : n \geq 1, m \geq 1\}$. Then $L_1 \subseteq L_2$ since $w = a^n b^n \in L_1$ implies $w \in L_2$, just taking $m = n$. Since $L_1 \subseteq L_2$, we get $L_1 \cup L_2 = L_2$. Thus, this problem is

equivalent to determining if L_2 is regular.

L_2 is regular since it can be denoted by the regular expression aa^*bb^* , that is at least one a followed by at least one b . Thus, we conclude that $L = L_1 \cup L_2 = L_2$ is regular.

(b) $L = \{a^n b^n : n \geq 1\} \cup \{a^n b^{n+2} : n \geq 1\}$.

We will show L is not regular by using the pumping lemma.

- For a contradiction, assume L is regular.
- Since L is a regular there exists a pumping length $m > 0$.
- Take $w = a^m b^m$, then $w \in L$ and $|w| > m$.
- Then from the pumping lemma there exists $x, y, z \in \Sigma^*$ such that $w = xyz$ with (1) $|xy| \leq m$ and (2) $|y| \geq 1$.
- From (1) and (2) above, $y = a^k$ for some $1 \leq k \leq m$.
- Take $i = 2$, then $xy^i z = a^{m+k} b^m \notin L$ because the number of a 's is greater than the number of b 's, so $xy^i z$ cannot be in either part of the union.
- This is a contradiction of the pumping lemma. Thus, L cannot be regular.

4.3.7: Show that the language

$$L = \{a^n b^n : n \geq 0\} \cup \{a^n b^{n+1} : n \geq 0\} \cup \{a^n b^{n+2} : n \geq 0\}$$

is not regular.

We will show L is not regular by using the pumping lemma.

- For a contradiction, assume L is regular.
- Since L is a regular there exists a pumping length $m > 0$.
- Take $w = a^m b^m$, then $w \in L$ and $|w| > m$.
- Then from the pumping lemma there exists $x, y, z \in \Sigma^*$ such that $w = xyz$ with (1) $|xy| \leq m$ and (2) $|y| \geq 1$.
- From (1) and (2) above, $y = a^k$ for some $1 \leq k \leq m$.
- Take $i = 2$, then $xy^i z = a^{m+k} b^m \notin L$ because the number of a 's is greater than the number of b 's, so $xy^i z$ cannot be in any of the three parts of the union.
- This is a contradiction of the pumping lemma. Thus, L cannot be regular.

Problem 3: Linz 4.1.16

Show that the statement “If L_1 is regular and $L_1 \cup L_2$ is also regular, then L_2 must be regular” were true for all L_1 and L_2 , then all languages would be regular.

Let $L_1 = \Sigma^*$, then L_1 is regular (denoted by the regular expression Σ^* , where the regular expression Σ is as defined in the solution of Problem 1, 4.1.13 above). Let L_2 be any language. By the definition of a language, $L_2 \subseteq L_1$. Therefore, $L_1 \cup L_2 = L_1$, and hence is regular. Thus, if the statement in the problem description were true, then L_2 must be regular. Since L_2 was any arbitrary language, we would have that all languages would be regular.

The statement in this problem is clearly false though. Let $L_1 = \Sigma^*$, $L_2 = \{a^n b^n : n \geq 0\}$. Then L_1 is regular, and $L_1 \cup L_2 = L_1$ is regular, but we know L_2 is not regular, disproving the statement.

Problem 4: Linz 4.2.14

Find an algorithm for determining whether a regular language L contains an infinite number of even-length strings.

On input L where L is a regular language:

1. Represent L as a DFA, D .
2. Construct a DFA for the language $L' = \{w : |w| \text{ is even}\}$ (L' is regular since it can be represented by the RE $(\Sigma\Sigma)^*$).
3. Construct a DFA for $L'' = L \cap L'$, which is regular from Theorem 4.1 . (This is the language of all even length strings in L).
4. Now we can follow the algorithm in Theorem 4.6 to determine if L'' is infinite, which will give us the result of whether L has an infinite number of even-length strings.

Note the above algorithm will return that L contains an infinite number of even-length strings iff L'' is infinite. This is true since L'' contains all the even-length strings in L , because it is the intersection of L and the language of all even-length strings, L' .

Problem 5: Linz 4.3.15 (c)-(g)

Consider the languages below. For each, make a conjecture whether or not it is regular. Then prove your conjecture.

(c) $L = \{a^n b^l : n/l \text{ is an integer}\}$.

We will show L is not regular by using the pumping lemma.

- For a contradiction, assume L is regular.
- Since L is a regular there exists a pumping length $m > 0$.
- Take $w = a^{m+1}b^{m+1}$, then $w \in L$ ($\frac{m+1}{m+1} = 1$) and $|w| > m$.

- Then from the pumping lemma there exists $x, y, z \in \Sigma^*$ such that $w = xyz$ with (1) $|xy| \leq m$ and (2) $|y| \geq 1$.
- From (1) and (2) above, $y = a^k$ for some $1 \leq k \leq m$.
- Take $i = 0$, then $xy^iz = a^{m+1-k}b^{m+1} \notin L$ because $0 < \frac{m+1-k}{m+1} < 1$, since $0 < m+1-k < m+1$.
- This is a contradiction of the pumping lemma. Thus, L cannot be regular.

(d) $L = \{a^n b^l : n + l \text{ is a prime number}\}$.

We will show L is not regular by using a closure property and then the pumping lemma.

- For a contradiction, assume L is regular.
- Then from Theorem 4.3 $h(L)$ is regular for any homomorphism h .
- Let h be the homomorphism defined as $h(a) = a$ and $h(b) = a$.
- Then $h(L) = \{a^n a^l : n + l \text{ is a prime number}\} = \{a^{n+l} : n + l \text{ is a prime number}\} = \{a^p : p \text{ is a prime number}\}$.
- We will now show $h(L)$ cannot be regular causing the contradiction.
- Since $h(L)$ is a regular there exists a pumping length $m > 0$.
- Take $w = a^p$, where p is a prime number larger than m (Such a p exists since there is an infinite number of prime numbers). Then $w \in L$ and $|w| > m$.
- Then from the pumping lemma there exists $x, y, z \in \Sigma^*$ such that $w = xyz$ with (1) $|xy| \leq m$ and (2) $|y| \geq 1$.
- From (1) and (2) above, $y = a^k$ for some $1 \leq k \leq m$.
- Take $i = p + 1$, then $xy^iz = a^{p+kp} = a^{p(k+1)} \notin L$ because there are $p * (k + 1)$ a 's, where $p, k + 1 > 1$. Thus, the number of a 's cannot be prime.
- This is a contradiction of the pumping lemma. Thus $h(L)$ cannot be regular, which contradicts Theorem 4.3. Thus, we conclude L cannot be regular.

(e) $L = \{a^n b^l : n \leq l \leq 2n\}$.

We will show L is not regular by using the pumping lemma.

- For a contradiction, assume L is regular.
- Since L is a regular there exists a pumping length $m > 0$.
- Take $w = a^m b^m$, then $w \in L$ and $|w| > m$.
- Then from the pumping lemma there exists $x, y, z \in \Sigma^*$ such that $w = xyz$ with (1) $|xy| \leq m$ and (2) $|y| \geq 1$.

- From (1) and (2) above, $y = a^k$ for some $1 \leq k \leq m$.
- Take $i = 2$, then $xy^iz = a^{m+k}b^m \notin L$ because the number of a 's ($m+k$) is not less than or equal to the number of b 's (m) in xy^iz .
- This is a contradiction of the pumping lemma. Thus, L cannot be regular.

(f) $L = \{a^n b^l : n \geq 100, l \leq 100\}$.

L is regular. To see this, note $L = L_1 L_2$, where $L_1 = \{a^n : n \geq 100\}$ and $L_2 = \{b^l : l \leq 100\}$. We will show that L_1 and L_2 are regular.

Define an NFA $N_1 = (\{q_0, q_1, \dots, q_{100}\}, \{a, b\}, \delta_1, q_0, \{q_{100}\})$, where $\delta_1(q_i, a) = \{q_{i+1}\}$ for $0 \leq i \leq 99$, $\delta_1(q_{100}, a) = \{q_{100}\}$ and $\delta_1(q_i, b) = \{\}$ for $0 \leq i \leq 100$. Then, $L(N_1) = L_1$ since a string is accepted by N_1 iff it has at least 100 a 's and no b 's. Since there is an NFA that accepts L_1 , L_1 is regular.

Define an NFA $N_2 = (\{q_0, q_1, \dots, q_{100}\}, \{a, b\}, \delta_2, q_0, \{q_0, q_1, \dots, q_{100}\})$, where $\delta_2(q_i, b) = \{q_{i+1}\}$ for $0 \leq i \leq 99$, $\delta_2(q_{100}, b) = \{\}$ and $\delta_2(q_i, a) = \{\}$ for $0 \leq i \leq 100$. Then, $L(N_2) = L_2$ since a string is accepted by N_2 iff it has at most 100 b 's and no a 's. Since there is an NFA that accepts L_2 , L_2 is regular.

Since L_1 and L_2 are regular and regular languages are closed under concatenation, then $L = L_1 L_2$ is regular.

(g) $L = \{a^n b^l : |n - l| = 2\}$.

We will show L is not regular by using the pumping lemma.

- For a contradiction, assume L is regular.
- Since L is a regular there exists a pumping length $m > 0$.
- Take $w = a^m b^{m+2}$, then $w \in L$ and $|w| > m$.
- Then from the pumping lemma there exists $x, y, z \in \Sigma^*$ such that $w = xyz$ with (1) $|xy| \leq m$ and (2) $|y| \geq 1$.
- From (1) and (2) above, $y = a^k$ for some $1 \leq k \leq m$.
- Take $i = 0$, then $xy^iz = a^{m-k} b^{m+2} \notin L$ because $|m - k - (m + 2)| = |-k - 2| = k + 2 \neq 2$.
- This is a contradiction of the pumping lemma. Thus, L cannot be regular.